Analysis of Capacitive Coupling Configurations

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October 30, 2025

1 Analysis of Initial Circuit Configuration

This document presents an analysis of a capacitive coupling model, as depicted in Figure 1, which illustrates the various capacitive paths in an initial circuit setup.

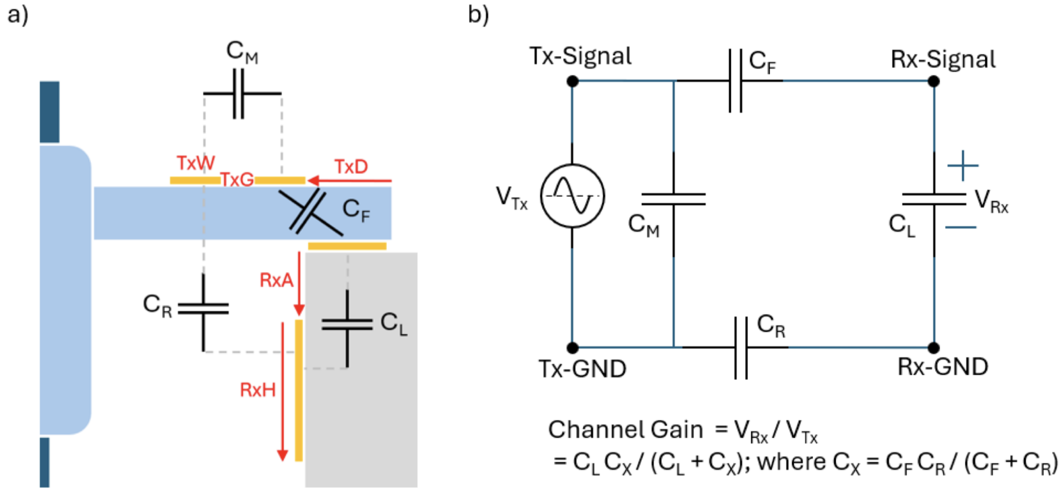

Figure 1: a) Physical representation of the capacitive coupling in an initial circuit setup. b) Equivalent circuit diagram showing the various capacitive components. *(Source: Screenshot 2025-10-30 at 3.02.37 PM)*

Component Definitions

Here is a breakdown of the capacitive model components:

1. C_M **(Mutual/Return-Path Capacitance):** Path represents the coupling between the two transmitter electrodes. Its physical interpretation is the capacitive coupling path directly through the human arm tissues (skin, fat, muscle, bone, etc.) between Tx-Signal and Tx-GND.

2. C_F **(Forward-Path Capacitance):** This is the capacitive coupling path from the Tx-Signal electrode to the Rx-Signal electrode. It models the signal coupling from the transmitting side to the receiving side through the body, air, and table. This is the primary forward path for the signal.

3. C_R **(Return-Path Capacitance):** This is the capacitive coupling path from the Tx-GND electrode to the Rx-GND electrode. It models the signal's return path to complete the circuit.

4. C_L **(Receiver Load Capacitance):** This represents the coupling between the two receiver electrodes. It is the capacitance of the load or the measurement device across which the voltage V_{Rx} is measured.

Simplified Derivation of Channel Gain

Note: This is an approximation based on a simplified model. A more rigorous derivation using nodal analysis is presented in Section 4.

For two capacitors in series, the equivalent capacitance (C_{eq}) is calculated as:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \implies C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

Applying this by modeling C_F and C_R as a series combination, we find their equivalent capacitance, C_X:

$$C_X = \frac{C_F C_R}{C_F + C_R} \quad \text{(Equivalent coupling capacitance)}$$

Applying the Voltage Divider Rule, the output voltage (V_{Rx}) is given by:

$$V_{Rx} = V_{Tx} \times \frac{Z_{C_L}}{Z_{C_X} + Z_{C_L}}$$

where the impedance of a capacitor is $Z_C = \frac{1}{j\omega C}$. Substituting the impedance terms into the equation:

$$V_{Rx} = V_{Tx} \times \frac{\frac{1}{j\omega C_L}}{\frac{1}{j\omega C_X} + \frac{1}{j\omega C_L}}$$

To simplify, we multiply the numerator and denominator by $j\omega C_L C_X$:

$$V_{Rx} = V_{Tx} \times \frac{C_X}{C_L + C_X}$$

Finally, rearranging for the channel gain (Gain $= V_{Rx}/V_{Tx}$):

$$\text{Channel Gain} = \frac{V_{Rx}}{V_{Tx}} = \frac{C_X}{C_L + C_X}$$

2 Analysis of Configuration 1

A second physical arrangement, "Config 1," is shown in Figure 2. While the physical setup changes, it can be represented by the same general electrical schematic as the initial configuration.

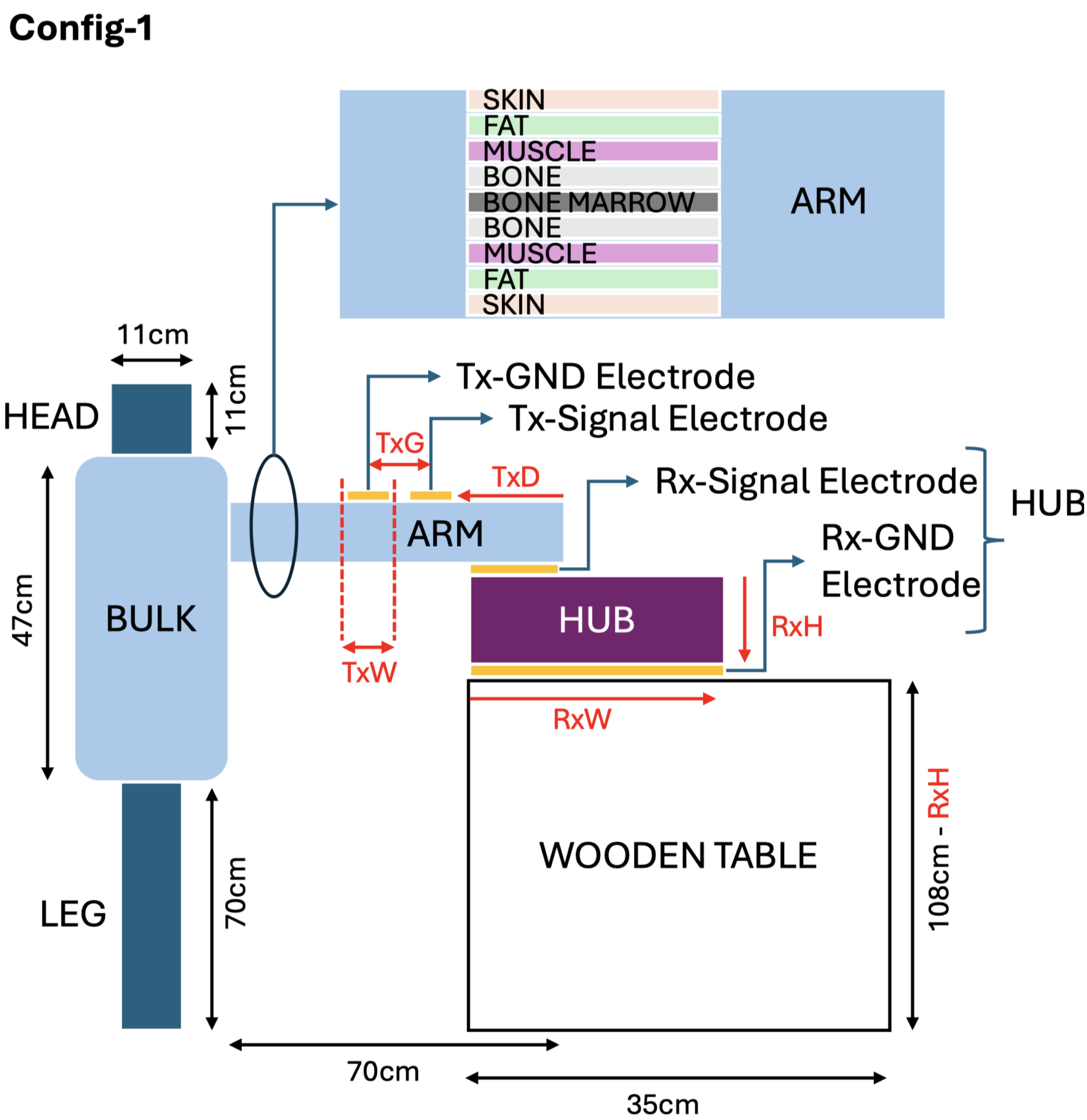

Figure 2: Physical representation and equivalent circuit for "Config 1". *(Source: Screenshot 2025-10-30 at 3.07.00 PM)*

3 Comparison of Circuit Models (Initial vs. Config 1)

Are the circuits for the initial setup and "Config 1" exactly the same? The answer is both yes and no.

3.1 Yes: The Schematic is a General Model

The fundamental circuit diagram (the topology) is the same for both scenarios. It is a general model for a four-terminal capacitive sensor and is valid for both configurations.

3.2 No: The Physical Reality and Component Values Differ

The difference lies in what each capacitor physically represents in each setup, which in turn dramatically changes their capacitance values. A change in values means a change in the circuit's overall behavior.

4 Detailed Derivation via Nodal Analysis (Initial & Config 1)

We will use Nodal Analysis on the equivalent circuit in the frequency domain to derive the accurate gain formula for the topology shown in Figures 1 and 2. We define the Tx-GND node as our reference ground (0V).

Let the voltage at the Rx-Signal node be V_1.
Let the voltage at the Rx-GND node be V_2.
Therefore, the output voltage is $V_{Rx} = V_1 - V_2$.
The voltage at the Tx-Signal node is fixed by the source at V_{Tx}.

Applying Kirchhoff's Current Law (KCL)

At Node V_1 (Rx-Signal):

$$\frac{V_1 - V_{Tx}}{Z_{C_F}} + \frac{V_1 - V_2}{Z_{C_L}} = 0 \implies V_1(C_F + C_L) - V_2 C_L = V_{Tx} C_F \quad \cdots \quad (1)$$

At Node V_2 (Rx-GND):

$$\frac{V_2 - V_1}{Z_{C_L}} + \frac{V_2 - 0}{Z_{C_R}} = 0 \implies V_2 = V_1 \frac{C_L}{C_L + C_R} \quad \cdots \quad (2)$$

Solving for the Gain

Substitute Eq. 4 into Eq. 4 to solve for V_1:

$$V_1 = V_{Tx} \frac{C_F(C_L + C_R)}{C_F C_L + C_F C_R + C_L C_R}$$

Now, find the output voltage, $V_{Rx} = V_1 - V_2 = V_1 \left(\frac{C_R}{C_L + C_R}\right)$. Substituting the expression for V_1:

$$V_{Rx} = \left(V_{Tx} \frac{C_F(C_L + C_R)}{C_F C_L + C_F C_R + C_L C_R}\right) \left(\frac{C_R}{C_L + C_R}\right)$$

The $(C_L + C_R)$ terms cancel, leaving the final expression:

$$\text{Gain} = \frac{V_{Rx}}{V_{Tx}} = \frac{C_F C_R}{C_F C_L + C_F C_R + C_L C_R}$$

5 Analysis of Configuration 2

A third configuration, "Config-2," utilizes a different physical arrangement and results in a simpler equivalent circuit model, shown in Figures 3 and 4.

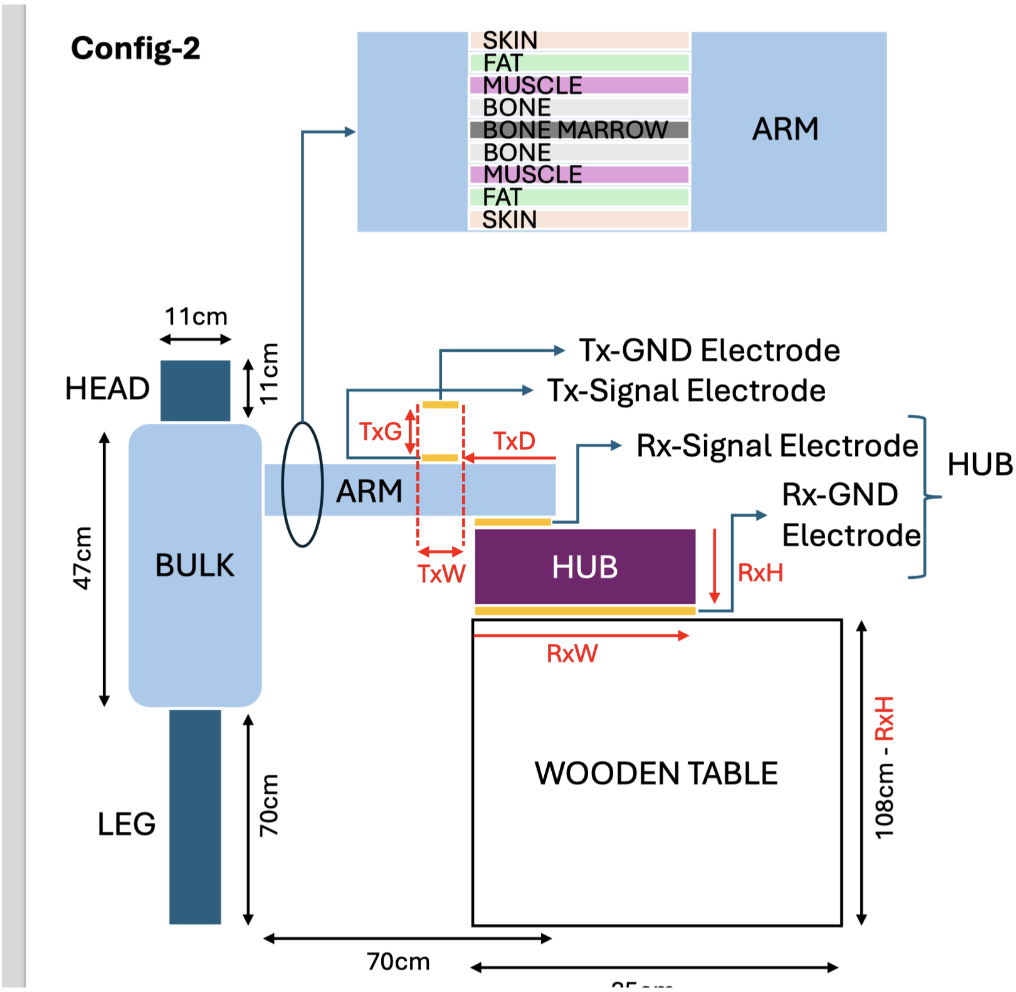

Figure 3: Physical model for Configuration 2. *(Source: Screenshot 2025-10-30 at 4.33.08 PM)*

5.1 Circuit Simplification

First, let's simplify the circuit based on two key observations:

1. **Parallel Capacitors (C_{TR} and C_{AR}):** The capacitors C_{TR} and C_{AR} are connected in parallel between the "Tx-Signal / Arm" node and the "Rx-Signal" node. We can combine them into a single equivalent capacitance, C_{eq}.
$$C_{eq} = C_{TR} + C_{AR}$$

2. **Capacitor in Parallel with Source (C_{AG}):** The capacitor C_{AG} is connected directly across the ideal voltage source V_{Tx}. A component in parallel with an ideal voltage source does not affect the voltage supplied to the rest of the circuit. Since we are calculating a voltage ratio (V_{Rx}/V_{Tx}), C_{AG} has no effect on the gain.

After these simplifications, we are left with a very simple series circuit, which is a classic capacitive voltage divider.

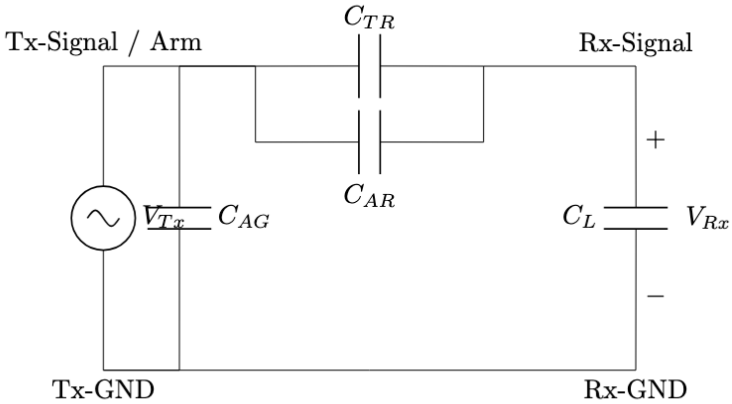

Figure 1: Simplified equivalent circuit for Config-2 based on the FEMM simulation assumption (Tx-Signal and Arm are one node).

Figure 4: Equivalent circuit for Configuration 2 based on FEMM analysis. *(Source: Screenshot 2025-10-30 at 6.27.40 PM)*

5.2 Gain Calculation using the Voltage Divider Rule

We start with the voltage divider formula using impedances:

$$V_{Rx} = V_{Tx} \cdot \frac{Z_{C_L}}{Z_{C_{eq}} + Z_{C_L}}$$

Step 1: Substitute the impedance values.

$$V_{Rx} = V_{Tx} \cdot \frac{\frac{1}{j\omega C_L}}{\frac{1}{j\omega C_{eq}} + \frac{1}{j\omega C_L}}$$

Step 2: Simplify the complex fraction. A clear way to simplify this is to multiply the numerator and the denominator of the large fraction by $j\omega$. This cancels out all the $j\omega$ terms.

$$V_{Rx} = V_{Tx} \cdot \frac{\frac{1}{j\omega C_L} \cdot (j\omega)}{(\frac{1}{j\omega C_{eq}} + \frac{1}{j\omega C_L}) \cdot (j\omega)} = V_{Tx} \cdot \frac{\frac{1}{C_L}}{\frac{1}{C_{eq}} + \frac{1}{C_L}}$$

Step 3: Simplify the remaining fraction. Now we have a simpler fraction. To get rid of the fractions in the denominator, we multiply the numerator and denominator of the large fraction by the common term $C_{eq}C_L$.

$$V_{Rx} = V_{Tx} \cdot \frac{\frac{1}{C_L} \cdot (C_{eq}C_L)}{(\frac{1}{C_{eq}} + \frac{1}{C_L}) \cdot (C_{eq}C_L)}$$

Evaluating the top and bottom parts separately:

- Numerator: $\frac{1}{C_L} \cdot (C_{eq}C_L) = C_{eq}$
- Denominator: $(\frac{1}{C_{eq}} \cdot C_{eq}C_L) + (\frac{1}{C_L} \cdot C_{eq}C_L) = C_L + C_{eq}$

Step 4: Combine to get the final result. Substituting these back into the expression gives us the final, correct relationship:

$$V_{Rx} = V_{Tx} \cdot \frac{C_{eq}}{C_{eq} + C_L}$$

5.3 The Final Gain Formula for Config-2

Finally, we substitute $C_{eq} = C_{TR} + C_{AR}$ back into the equation. The channel gain $G = V_{Rx}/V_{Tx}$ for this circuit is:

$$\text{Gain} = \frac{V_{Rx}}{V_{Tx}} = \frac{C_{TR} + C_{AR}}{C_{TR} + C_{AR} + C_L}$$

Analysis of the Result:

- **Dimensionless:** The formula is a ratio of capacitances (Farads / Farads), so it is correctly dimensionless.
- **Intuitive:** The gain is determined by the ratio of the total coupling capacitance ($C_{TR} + C_{AR}$) to the total capacitance in the series path ($C_{TR} + C_{AR} + C_L$). As the coupling to the receiver increases or the load capacitance decreases, the gain increases, which makes physical sense.